Polynomial-Resource Classification of Quantum Circuit Families via Classical Shadows

Andrew Maciejunes
Virginia Modeling, Analysis and Simulation Center
Old Dominion University
Norfolk, VA, USA
amacieju@odu.edu

Ross Gore
Center for Secure and Intelligent Critical Systems
Old Dominion University
Norfolk, VA, USA
rgore@odu.edu

***Abstract*—We demonstrate that three structurally distinct quantum circuit families: Instantaneous Quantum Polynomial (IQP), Clifford, and Clifford+T, can be reliably distinguished using only polynomial measurement resources. The central methodological contribution is the application of classical shadows to extract all six two-qubit Pauli correlators (ZZ, XX, YY, XY, XZ, YZ) across all qubit pairs, yielding $O(n^2)$ features from a shot budget that scales as $s = 16n^2$—entirely polynomial in system size. We compare three measurement strategies: a Z-basis-only baseline (all-pairs ZZ correlations), a multi-basis approach (global Z/X/Y rotations giving ZZ, XX, and YY correlators at $3\times$ the shot cost), and classical shadows (all six Pauli pair correlators at the same shot cost as the baseline). Four classical classifiers are evaluated across 4–20 qubits with $N_c = 1000$ circuits per family. [TODO: Summarize key accuracy result and scaling trend.]**

***Index Terms*—classical shadows, quantum circuit families, IQP circuits, Clifford circuits, polynomial resources, distribution classification, Pauli correlators, quantum machine learning**

I. Introduction

A fundamental question in quantum information is whether the output distributions of quantum circuits, distributions that cannot be efficiently represented classically, can be meaningfully characterized using only polynomially many measurements. Quantum state tomography, originally proposed by Vogel and Risken (1989) [1] and first demonstrated experimentally by Smithey et al. (1993) [2], provides a complete reconstruction of the quantum state but at prohibitive cost. Haah et al. (2016) [3] rigorously established this cost with a lower bound of $\Omega(d^2/\delta)$ copies, where d is the Hilbert space dimension and δ is the desired precision. Since $d = 2^n$ for an n-qubit system, full tomography requires exponentially many measurements in the number of qubits, rendering it infeasible for systems of practical interest. This exponential barrier motivates the search for protocols that extract meaningful information about quantum states without full reconstruction.

Quantum circuit families provide a principled testbed for this question. IQP, Clifford, and Clifford+T circuits produce distributions with qualitatively different statistical signatures, rooted in their distinct computational complexity properties. Clifford circuits are classically simulable; IQP sampling is believed to be classically hard under plausible complexity-theoretic conjectures; Clifford+T circuits interpolate between the two. If these families can be distinguished with polynomially many measurements, it suggests that their distributional signatures are encoded in low-order statistics that a polynomial measurement protocol can access.

The key methodological challenge is choosing the right measurement protocol. Measuring in a single basis (Z-only) gives access to ZZ correlators but misses the X and Y components of the Pauli correlation structure. Measuring in three global bases (Z, X, Y) recovers ZZ, XX, and YY correlators but requires $3\times$ the shot budget and still misses mixed-basis correlators (XY, XZ, YZ). Classical shadowsresolve this: by applying a random single-qubit Clifford rotation to each qubit before measurement, all six two-qubit Pauli correlators can be estimated from a single unified measurement protocol, with sample complexity $O(n^2)$—matching the shot budget required for any individual correlator type.

Contributions:

- A systematic comparison of three polynomial measurement strategies (Z-only, multi-basis, classical shadows) for classifying quantum circuit families across 4–20 qubits.
- A classical shadows pipeline for quantum circuit classification that extracts all six two-qubit Pauli correlator types (ZZ, XX, YY, XY, XZ, YZ) across all pairs at $O(n^2)$ shot cost, yielding $3n^2$ features.
- An analysis of the shot requirements of candidate features, motivating the exclusion of Shannon entropy and collision probability and the adoption of a quadratic shot scaling law $s = 16n^2$.

II. Background

A. Circuit Families as Distribution Classes

We consider three quantum circuit families that span a range of computational complexity.

a) Clifford circuits: Generated by $\{H, S, \mathrm{CNOT}\}$, Clifford circuits are efficiently simulable classically via the stabilizer formalism.Their output distributions possess structured pairwise correlations reflecting the underlying stabilizer group geometry.

b) Clifford+T circuits: Adding $T = \mathrm{diag}(1, e^{i\pi/4})$ yields a universal gate set.Sufficient T-gate density breaks the stabilizer structure and moves the circuit outside the efficiently

simulable regime, producing a distributional signature intermediate between Clifford and IQP.

c) IQP circuits: Instantaneous Quantum Polynomial circuits apply diagonal gates in the computational basis, sandwiched by Hadamard layers.Classical hardness of IQP sampling is supported by complexity-theoretic arguments, and their parity structure produces a distributional fingerprint qualitatively distinct from the Clifford family.

B. Classical Shadows

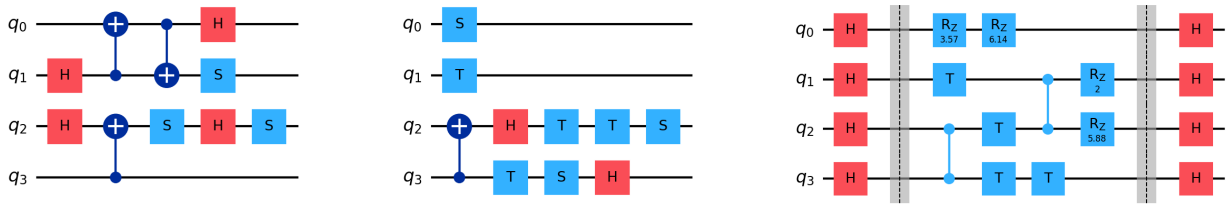
Classical shadowsprovide an efficient protocol for estimating many properties of a quantum state from a single randomized measurement dataset. For each measurement shot, a random single-qubit unitary U_i is applied to each qubit i before measurement in the computational basis, and the rotation choice is recorded alongside the outcome. The measurement channel can then be inverted classically to produce unbiased estimators for arbitrary observables.

For a single-qubit Pauli $P_i \in \{X, Y, Z\}$, the shadow estimator is:

$$\hat{P}_i = 3 \cdot (-1)^{b_i} \cdot \mathbf{1}[U_i \text{ matches } P_i\text{'s basis}], \tag{1}$$

where the factor of 3 corrects for the $1/3$ probability of any given basis being selected. For a two-qubit correlator $P_i P_j$, the estimator is the product of single-qubit estimators, yielding a factor of 9 and requiring both U_i and U_j to match their respective Pauli bases.

The connected correlator $\langle P_i P_j \rangle_c = \langle P_i P_j \rangle - \langle P_i \rangle \langle P_j \rangle$ can be estimated by combining single- and two-qubit shadow estimators. For K Pauli operators, the required number of shadows scales as $O(K/\varepsilon^2)$ for bounded observables, independent of system size.For all $6 \cdot \binom{n}{2}$ two-qubit Pauli correlators, this yields a total shot cost of $O(n^2/\varepsilon^2)$—polynomial in n.



(a) Clifford (b) Clifford+T (c) IQP

Fig. 1. We illustrate an example Clifford, Clifford+T, and IQP circuit for the reader with four qubits and N_c = 10. (a) The Clifford circuit is comprised of gates from the clifford gate as defined above. (b) We observe a disproportionate amount of T gates for the Clifford+T example. This is done to ensure distinctness from the Clifford circuit family. (c) We show an example IQP circuit. It is clear that N_c for IQP circuits refers to the number of operations between the Hadamard gates.

C. Distribution Classification

[TODO: Brief review of relevant prior work on distribution testing, quantum state discrimination, and classical analogs. Position our contribution relative to these.]

III. Methods

A. Circuit Generation

We use the Qiskit SDK [4] for all of our research. Using Qiskit, we craft random circuits for each of the three previously mentioned circuit families, Clifford, Clifford+T, and IQP circuits. For the first two groups, we define a value N_c, which is the number of gates in a given circuit. We then apply N_c randomly selected gates from the appropriate circuit family to act on randomly selected qubits. For the Clifford family of circuits, each gate has an equal chance of being selected. For the Clifford+T family, the gates have chances of being selected as outlined in TableIII-A. The qubit(s) each gate acts on is selected randomly with no preference to any particular qubit, except in the case of two qubit gates, where the control and target must be different.

TABLE I
Clifford+T gate selection probabilities. All other families used equal probabilities.

Gate	Probability
S	0.2
CNOT	0.2
H	0.2
T	0.4

Instantaneous Quantum Polynomial (IQP) circuits are created simiiarly. The only difference, beside the gate set used, is that the operator $H^{\otimes n}$ is applied at the beginning and end of the circuit, as shown in Figure 1, where n is the number of qubits. We scale the number of qubits from 4 up to 20 qubits. At each qubit count, we generate 1000 circuits of each type.

B. Shot Budget Design

The estimation cost of a statistical feature determines the minimum shot budget needed to estimate it reliably.

TABLE II
Experimental parameter sweep.

Parameter	Values
Circuit families	IQP, Clifford, Clifford+T
Qubit counts	4–20
Shot budget	$16n^2$ (quadratic)
Circuits per family per qubit count	1000
Operations per circuit	1000
Simulator	Qiskit Aer (statevector, GPU)
Measurement strategies	Z-only, multi-basis, shadows

a) Polynomial-cost features: The Hamming-weight histogram bins all 2^n outcomes into $n+1$ weight classes, requiring $O(n/\varepsilon^2)$ shots. Single-qubit marginals $P(x_i = 1)$ and parity bias require $O(n/\varepsilon^2)$ and $O(1/\varepsilon^2)$ shots respectively. Connected two-qubit Pauli correlations across all $\binom{n}{2}$ pairs require $O(n^2/\varepsilon^2)$ shots in total for any single correlator type (e.g., ZZ, XX, YY). By the classical shadows result, *all* $6\binom{n}{2}$ two-qubit Pauli correlators (ZZ, XX, YY, XY, XZ, YZ) can be estimated simultaneously at the same $O(n^2/\varepsilon^2)$ cost using the shadow protocol.

b) Excluded exponential-cost features: The purpose of this work is to classify quantum circuit families using only polynomial measurement resources. If exponential resources were available, full quantum state tomography could exactly reconstruct the output distribution, rendering classification trivial. In that vein, several prominent statistical measures are excluded from our feature set due to their exponential shot requirements: Shannon entropy $H = -\sum_x \hat{p}(x)\log_2 \hat{p}(x)$, which requires $O(2^n/n)$ shots because the empirical frequency estimator is biased whenever bitstrings go unobserved; collision probability $\mathcal{C} = \sum_x p(x)^2$, which requires $O(2^{n/2})$ shots via the birthday problem threshold; Rényi entropies of order $\alpha \neq 1$, which share the same exponential sampling barrier; and total variation distance to uniformity, which requires observing a representative fraction of the 2^n outcomes. All such features are incompatible with the polynomial-resource constraint central to this work and are therefore disregarded.

c) Quadratic shot scaling: The most expensive retained features are the two-qubit Pauli correlators, requiring $O(n^2)$ shots. We adopt a quadratic shot scaling law $s = \lambda n^2$, where

s is the total shot budget, n is the number of qubits, and λ is a constant prefactor controlling the statistical quality of the estimators. We arbitrarily set $\lambda = 16$, giving 256 shots at $n = 4$ and 6400 shots at $n = 20$. For the multi-basis strategy, each of the three basis runs receives $s/3$ shots; the total shot budget remains $s = \lambda n^2$.

C. Measurement Strategies

We compare three measurement protocols, each operating within the same $s = 16n^2$ total shot budget.

a) Z-only baseline: All shots are taken in the computational basis. Features extracted: Hamming-weight histogram, Z-basis single-qubit marginals, parity bias, and all-pairs connected ZZ correlations. This is the minimal polynomial-resource strategy and serves as the baseline.

b) Multi-basis: Shots are divided equally among three global basis rotations: identity (Z basis), Hadamard on all qubits (X basis), and $S^\dagger H$ on all qubits (Y basis), with $s/3$ shots per basis. Features extracted: Hamming-weight histogram, $Z/X/Y$ single-qubit marginals, parity bias, and all-pairs connected ZZ, XX, and YY correlations. This gives three of the six Pauli pair correlator types at the same total shot cost, at the expense of $3\times$ fewer shots per basis run. Mixed-basis correlators (XY, XZ, YZ) are not accessible with global rotations.

c) Classical shadows: For each of the s measurement shots, a random basis (Z, X, or Y) is independently selected for each qubit, and the circuit is measured after applying the corresponding single-qubit rotation. All six two-qubit Pauli correlator types (ZZ, XX, YY, XY, XZ, YZ) are estimated from the resulting shadow dataset using the unbiased shadow estimators described in Section II. This achieves the full Pauli correlation structure at the same total shot budget as the Z-only baseline.

D. Feature Extraction

a) Z-only features: The feature vector concatenates: Hamming-weight histogram ($n + 1$ values), Z-marginals (n values), parity bias (1 value), and all-pairs ZZ connected correlations ($\binom{n}{2}$ values):

$$d_{\mathrm{Z}} = (n + 1) + n + 1 + \tfrac{n(n-1)}{2} = \tfrac{n^2+3n+4}{2}. \tag{2}$$

This grows as $O(n^2)$, dominated by the all-pairs ZZ term.

b) Multi-basis features: The feature vector extends the Z-only set by appending X-marginals and all-pairs XX correlations, then Y-marginals and all-pairs YY correlations:

$$d_{\mathrm{MB}} = d_{\mathrm{Z}} + 2n + n(n-1) = O(n^2). \tag{3}$$

c) Shadow features: The feature vector concatenates $Z/X/Y$ single-qubit expectations ($3n$ values) and all six Pauli pair connected correlations across all $\binom{n}{2}$ pairs ($6 \cdot \binom{n}{2}$ values):

$$d_{\mathrm{S}} = 3n + 6 \cdot \tfrac{n(n-1)}{2} = 3n^2. \tag{4}$$

All three feature vectors scale as $O(n^2)$, consistent with the polynomial-resource design goal.

E. Classification Pipeline

We evaluate four classifiers from scikit-learn: Logistic Regression (L-BFGS, `max_iter`=1000), Decision Tree (unlimited depth), Random Forest (100 estimators), and Support Vector Machine (RBF kernel). Data are split 80/20 train/test. Accuracy is reported as the fraction of correctly classified test samples. Each classifier is independently trained and evaluated for each measurement strategy and qubit count.

IV. Results

A. Measurement Strategy Comparison

[TODO: Three line plots (one per strategy) of accuracy vs. qubit count, with one curve per classifier. Does shadows outperform Z-only at high qubit counts? Does multi-basis benefit from the additional correlator types, or does the $3\times$ shot dilution hurt?]

B. Classification Accuracy vs. System Size

Figure **??** shows average accuracy as a function of qubit count for each measurement strategy.

[TODO: Describe scaling trend. Does accuracy degrade, plateau, or improve with qubit count? Is there a crossover where shadows outperforms multi-basis and Z-only?]

C. Per-Family Confusion Analysis

[TODO: Include confusion matrix or per-class precision/recall. Which pairs of families are hardest to distinguish? Does shadows improve separation of the hardest pair?]

V. Discussion

A. Classical Shadows as the Right Measurement Protocol

The central finding of this work is that classical shadows enables polynomial-resource classification of exponentially large quantum distributions. The key insight is that the information distinguishing IQP, Clifford, and Clifford+T families is encoded in low-order Pauli correlations—quantities that classical shadows can estimate efficiently. The full probability distribution, which would require exponential resources to characterize, is not needed.

The comparison between multi-basis and shadows is instructive. Multi-basis measurements access three of the six Pauli correlator types, but at the cost of $3\times$ shot dilution per basis run. Classical shadows access all six types at no additional shot cost, by using random per-qubit basis selection rather than fixed global rotations. Whether the additional correlator types (XY, XZ, YZ) provide sufficient discriminative power to compensate for the inherently noisier per-qubit basis estimates is an empirical question that our experimental comparison directly addresses.

B. Limitations

This study uses noiseless classical simulation. Real device noise will alter the distributions and likely degrade classification accuracy. Extending to noisy simulation (Aer noise models) or real hardware is a natural next step.

The shot budget $s = 16n^2$ is chosen to maintain consistent statistical quality across qubit counts. The constant factor of 16 is heuristic; a principled choice requires knowing the signal-to-noise ratio of the class-separating Pauli correlators, which is circuit-family-dependent.

C. Path to Broader Application

[TODO: Discuss what it would take to apply this pipeline to a concrete real-world domain: identify what quantity maps to system dimension and sample count, establish which distribution class the process belongs to, generate labeled circuit data, deploy the linear classifier.]

VI. Conclusion

We have demonstrated that classical shadows provides a polynomial-resource measurement protocol sufficient to distinguish IQP, Clifford, and Clifford+T circuit families across 4–20 qubits. By accessing all six two-qubit Pauli correlator types (ZZ, XX, YY, XY, XZ, YZ) across all qubit pairs at $O(n^2)$ shot cost, classical shadows yields an $O(n^2)$-dimensional feature vector that captures the distributional signatures of these families without requiring exponential measurement resources.

A systematic comparison with a Z-only baseline and a multi-basis strategy isolates the contribution of the full Pauli correlation structure and the effect of shot budget allocation. The quadratic shot budget $s = 16n^2$ is sufficient for all three strategies, confirming that the polynomial-resource design is consistent across the 4–20 qubit range.

[TODO: State best accuracy achieved and at what resource level.]

Acknowledgments

[TODO: Funding sources, HPC allocation, institutional acknowledgments.]

References

[1] K. Vogel and H. Risken, "Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase," *Physical Review A*, vol. 40, no. 5, p. 2847, 1989.

[2] D. T. Smithey, M. Beck, M. G. Raymer, and A. Faridani, "Measurement of the wigner distribution and the density matrix of a light mode using optical homodyne tomography: Application to squeezed states and the vacuum," *Physical review letters*, vol. 70, no. 9, p. 1244, 1993.

[3] J. Haah, A. W. Harrow, Z. Ji, X. Wu, and N. Yu, "Sample-optimal tomography of quantum states," in *Proceedings of the forty-eighth annual ACM symposium on Theory of Computing*, 2016, pp. 913–925.

[4] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, "Quantum computing with Qiskit," 2024.